Fundamentals 23

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Problem 23.1

Part 1)

Let $f : X_1 \times X_2 \to X_2 \times X_1$ be defined as $f((a, b)) = (b, a)$. The inverse is f, so it is a homeomorphism.

Part 2)

Define $X_1 \times \cdots \times X_n = (X_1 \times \cdots \times X_{n-1}) \times X_n$.

Part 3)

Define $X_1 \times \cdots \times X_n$ as the set of all $(x_1, \ldots, x_n)$ where $x_1, \ldots, x_n \in X_1, \ldots, X_n$.

Problem 23.3

We need to show that $f^{-1}((x, x)) = x$ is continuous. Let $V = \{(x_\alpha, x_\alpha) : \alpha \in A\}$ be open. Since $V \subseteq X \times X$ and V is open, it is a union of several $U_\beta \times U_\beta$ where U_β are open in X for all $\beta \in B$. So $f^{-1}(V) = \{x_\alpha : \alpha \in A\}$ is open in X, so f is continuous.

Problem 23.4

Consider the interval $(a, b) \subseteq X$, and the mapping $f : (a, b) \to \mathbb{R}^2$ to a figure-eight.

Problem 23.5

We can roll the square into a cylinder and connect one end to the inside of the other, creating a Klein bottle.

Problem 23.6

Reflexivity: Clearly $x = x$ so $x \sim x$.
Symmetry: If $x \sim y$ then either $x = y$, in which case $y = x$ so $y \sim x$, or $\{x, y\} = \{0, 1\}$ which is a symmetric condition.

Transitivity: Suppose $x \sim y$ and $y \sim z$. If $x = y$ and $y = z$ then $x = z$ so $x \sim z$. If $x = y$ but $y \neq z$, then $\{y, z\} = \{0, 1\}$, so $\{x, z\} = \{y, z\} = \{0, 1\}$ so $x \sim z$. Similarly if $x \neq y$ but $y = z$ then $x \sim z$. Finally, if $x \neq y$ and $y \neq z$ then $\{x, y\} = \{y, z\} = \{0, 1\}$, and since $x \neq y$, y is either 0 or 1 so $\{z\} = \{0, 1\} - \{y\} = \{x\}$, so $x \sim z$.

Problem 23.9

Let $X = \{a, b, c\}$ and

$$T_1 = \{\varnothing, \{a\}, \{a, b\}, \{a, c\}, X\},$$

$$T_2 = \{\varnothing, \{a\}, \{b\}, \{a, b\}, X\}.$$

We have that $|T_1| = |T_2| = 5$, but they are not homeomorphic.